

Section A: MCQs, Short Answers, and Fill In the Blanks [Marks: 30]

1. Which of the following is FALSE? [2 marks] [CO1]
 - A. Involving users in the design process ensures surprises for the user when the final product is delivered.
 - B. Involving users in the design process ensures timely training of the users, thus facilitating the ease of use of the final product.
 - C. Involving users in the design process promotes their sense of ownership.
 - D. Involving users in the design process makes them forgive minor errors and mistakes in the system.
2. Which of the following is TRUE for distributed cognition? [2 marks] [CO1, CO2]
 - A. Cognitive processes are only related to digital applications and systems.
 - B. Cognitive processes are solely dependent on non-verbal communication.
 - C. Cognitive processes are spread across individuals, artifacts, and internal and external representations.
 - D. Cognitive processes are confined to an individual's mind.
3. Consider the following scenarios: [Multiple Select Question] [2 marks] [CO1, CO2]
 - (I) In an online meeting, you learn the facial features of your colleagues.
 - (II) In an online tutorial, you learn how to use Figma professionally.

Which of the following are TRUE?

 - A. (I) is an example of intentional learning.
 - B. (I) is an example of incidental learning.
 - C. (II) is an example of intentional learning.
 - D. (II) is an example of incidental learning.
4. In the double diamond model of design, which phase follows the Discover phase? [2 marks] [CO1]
 - A. Deliver
 - B. Develop
 - C. Determine
 - D. Define
5. Which of the following is concerned with explaining our interaction with representations like maps, notes, and diagrams? [2 marks] [CO1, CO2]
 - A. Embodied cognition
 - B. External cognition
 - C. Focused attention
 - D. Divided attention
6. Which of the following designs promotes perception in an innovative messaging application? [2 marks] [CO1, CO2]
 - A. Distinct message tone for group chats vs individual chats
 - B. Displaying names of the sender in colors that contrast with the background color.
 - C. Neither (a) nor (b).
 - D. Both (a) and (b).
7. Write in one line how food delivery applications are meeting the following user experience goals: [2 marks] [CO2]
 - A. Rewarding
 - B. Motivating
8. Name the design principle being used in each the following scenarios: [6 marks] [CO2]
 - a. A mouse does a 'click' sound when clicked right or left.
 - b. Relative ordering of keys in a keyboard is same across different laptop manufacturers.
 - c. File option is placed at the top left of the toolbar in Microsoft Office.
 - d. A mixer grinder does not start unless the mixer jar is locked to the motor coupler of the machine.
 - e. Using a mouse as a paperweight.
 - f. Users cannot move the cursor beyond the physical space on the screen.
9. Which of the following are the problems with interface metaphors [2 marks] [CO2]:
 - a. Makes learning new systems difficult

- b. Can conflict with design principles
- c. Designers can inadvertently use bad existing designs and transfer the bad parts over
- d. Familiar artifact or actions helps people understand the underlying conceptual model

10. Which of the following is not the correct combination? [1 mark] [CO1].

- a. Explanatory theory – explicating relationships and processes
- b. predictive theory – testing hypotheses about user performance
- c. generative theory – creating something new
- d. descriptive theory - providing guidance for design

11. HCI designers use _____ organization to arrange the options in the toolbars of any UI. [1 mark][CO1]

12. The warning label in the following application (Refer to Fig 1.) is an example of _____. [2 marks] [CO1, CO2].

- a. Heavy Multitasking
- b. Digital Forgetting
- c. Exogenous control of attention
- d. Echoic Memory.

13. A food delivery application neither provides a clear confirmation nor allows the user to track their order after they have made the payment for their orders. This is an example of the gulf of _____. [2 marks]

14. The _____ Gestalt's principle states that we organize our perceptions into complete objects rather than as a series of parts. [1 mark] [CO1]

15. Color blindness is an example of _____ and _____ type of impairment. [2 marks] [CO1]

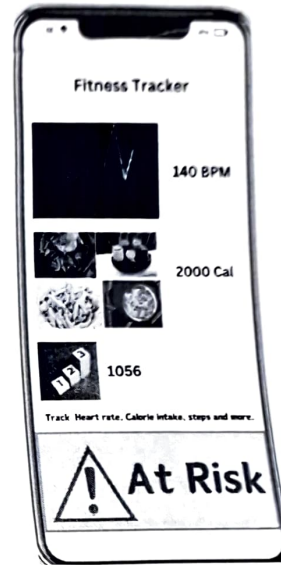


Fig. 1(Q12)

Section B: Subjective Questions [Marks: 20]

Explain the concept of an erroneous mental model using an example of an HCI application. Additionally, discuss at least three UX design methods that can improve users' mental models when interacting with this application. [Marks : 2+3] [CO1, CO2]

2. As a part of an Interaction Design team, you have been assigned to design and develop a pet care application that helps owners take better care of their pets. [Marks : 2+3] [CO1, CO2]
 - a. Briefly explain each phase of the simple interaction design lifecycle model with respect to the design of this application.
 - b. Discuss two design alternatives in terms of 1-2 features that you would like to design. The first design should focus on helping the users make better decisions about the choice of pet supplies (medicines, food, toys, accessories, etc.), and the second alternative should focus on making the user learn more about their pets.
3. In the context of HCI, what does Paradigm refer to? [2 marks, CO1]
4. Mention any two ways by which you think that the interface of MS Word promotes 'Exploration' type interaction? Justify your answer. [2 marks] [CO3]
5. Metro users can buy tickets for metro train either from the ticket counter or from the ticket kiosk. In the context of conceptual models, how do you think the acts of buying ticket from the kiosk and from the counter are similar. Mention atleast four similarities between the two. [4 marks] [CO3]
6. Give an example of: a) internal consistency; b) psychological constraint; c) accessible design
d) Audio Feedback [2 marks] [CO2]